Is Chain-of-Thought Just Rationalization?

A Mechanistic Faithfulness Audit of LLM Reasoning

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Project Overview

Chain-of-thought (CoT) prompting is the **default explanation interface** for LLMs in high-stakes settings: legal copilots, clinical decision support, and financial analysis tools. Human decision-makers re-calibrate trust based on these reasoning traces under the implicit assumption the chain *reflects* the computation that produced the answer.

The ethical problem: If CoT is post-hoc rationalization, a story constructed *after* the answer is decided, surfacing it to a clinician or judge actively **miscalibrates trust**, even when the final answer is correct.

> **Core Question:** Does a model's CoT causally mediate its answer, or is it rationalization written *after* the answer is decided?

We answer this by examining the model's **internal computations** using two interpretability tools applied to `meta-llama/Meta-Llama-3-8B-Instruct` (32 layers, fp16, A100) via TransformerLens.

Two experiments:

1. **Logit lens:** At each transformer layer, project the model's **internal representations** onto the output vocabulary and track when it commits to an answer class. Compare commitment depth under direct vs. CoT prompting.
2. **CoT corruption:** Replace the generated CoT with corrupted text (4 conditions of increasing meaning destruction) and measure how often the answer changes.

Datasets & Metrics

Dataset	Task	N
BoolQ	Yes/no reading comprehension	1500
MNLI	3-class NLI (entail/neutral/contra)	1000

Note: on MNLI, Llama-3 answers at near-chance accuracy (0.48, 3-way chance = 0.33) and usually produces an empty CoT, so MNLI served only as a failed generalization check. All results below are BoolQ.

Example (BoolQ):

Passage (excerpt)	*Question*
"Persian cats were developed in Iran, known for long fur..."	"Are Persian cats originally from Persia?"
Label: **Yes** Direct ℓ^*=29 With-CoT ℓ^*=30	

Metrics:

- **Commitment layer** ℓ^*: earliest layer where $P(\text{correct class}) \geq \tau$=0.8; reported as raw index and depth fraction ℓ^*/L
- **CoT Causal Influence Score (CIS):** fraction of examples where CoT corruption changes the predicted answer class
- **Faithfulness–accuracy correlation:** point-biserial r between flip indicator and clean-run accuracy. Positive r indicates genuine reasoning; negative r indicates a noise artifact

Methods & Experiments

Model: `meta-llama/Meta-Llama-3-8B-Instruct`, Llama-3 chat template, greedy decoding. Internal representations cached at every layer via `blocks.`ℓ`.hook_resid_post`.

Logit Lens: Commitment Detection

At each layer ℓ and final sequence position p:

$$\mathrm{LL}_\ell(p) \;=\; \mathrm{softmax}\big(W_U \cdot \mathrm{LN}_f(h_\ell^{(p)})\big), \quad h_\ell \in \mathbb{R}^d$$

Commitment layer: $\ell^*(x) = \min\{\ell : \mathrm{LL}_\ell[y] > \tau\}$, τ=0.8.

Run on two prompt variants per example:

- **Direct:** assistant turn ends with `"Answer:"` (no CoT)
- **With-CoT:** model generates a 60-token chain, then `"Answer:"` is appended

If CoT moves ℓ^* *earlier*, the answer was decided before the chain was written (rationalization). If *later*, the chain adds genuine elaboration.

CoT Corruption: 4-Condition Causal Test

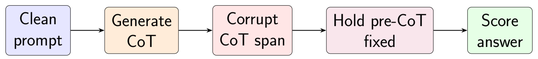
Let $[s, e)$ be the generated CoT token span. Four conditions ordered by how much meaning is destroyed:

Condition	What changes	n
Paraphrase	Nothing (meaning preserved)	400
Shuffle	Token order only	400
Random tokens	All content destroyed	800
Semantic invert	**Meaning inverted**	400

Since corruption only touches the CoT tokens, everything the model computed before the CoT is identical in both runs by construction. The flip rate (CIS) therefore directly measures how much the CoT content matters, and is compared across conditions.

A **purely post-hoc CoT** would show equal flip rates across all conditions. An ordering by how much meaning is destroyed is evidence of genuine causal influence.

Corruption pipeline:



References

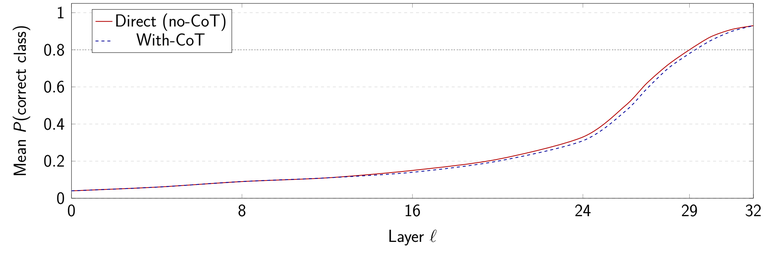
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Results

Logit Lens: Direct vs. CoT Commitment Depth (BoolQ, N=1500)

Condition	Mean ℓ^* [95% CI]	Depth frac.	Acc.
Direct (no-CoT)	29.42 [29.36, 29.48]	0.949	0.847
With-CoT	29.97 [29.95, 29.99]	0.967	0.838
Δ (CoT minus Direct)	**+0.55**	+0.018	−0.009

CoT moves commitment slightly *later* by 0.55 layers, not earlier. This is the opposite of what rationalization predicts. Accuracy is unchanged (overlapping CIs). Either way, generating a chain does not pull the decision forward.



CoT Corruption: Causal Influence Score (BoolQ)

Condition	CIS [95% CI]	
Paraphrase	0.043 [0.025, 0.062]	
Shuffle	0.085 [0.058, 0.113]	
Random	0.111 [0.090, 0.134]	
Sem. invert	**0.422** [0.375, 0.472]	$p<10^{-36}$ vs. paraphrase

The pattern is confirmed by accuracy: when the CoT is inverted, the model is more likely to flip on examples it was getting *right* (r=+0.10), a sign of genuine reasoning. Random corruption flips the uncertain cases (r=−0.21), a noise artifact.

Discussions & Future Research

Discussions:

> **Faithfulness is a matter of degree.**
>
> **(1)** The model locks in its answer in the **last 10% of layers** whether or not it generates a CoT ($\Delta\ell^*$=+0.55, slightly later not earlier).
>
> **(2)** The 4-condition corruption test **rules out pure rationalization** ($p\approx 4\times10^{-37}$, invert vs. paraphrase). The CoT does genuinely influence the answer.
>
> **(3)** But inversion only flips <50% of cases, so that influence is **real but limited**. The model partly commits before reading the chain.

Future Research:

- Layer-resolved patching to find where in the network CoT content enters the computation
- Replication across model families: Mistral-7B, Qwen2-7B, Gemma-2-9B
- Extension to legal and clinical reasoning tasks where trust miscalibration has the highest real-world stakes